

LIMA, Peru

WMO: 846280

Lat: 12.022S

Lon: 77.114W

Elev: 34

StdP: 100.91

Time Zone: -5.00 (W05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.0	14.6	11.9	8.7	15.7	12.1	8.8	15.7	7.8	18.7	7.2	18.2	1.9	180	0.337

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	6.2	29.0	23.0	27.9	22.5	27.0	22.1	23.7	27.3	23.1	26.6	22.6	26.0	4.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.3	17.1	26.1	21.9	16.7	25.8	21.2	15.9	25.0	71.3	27.2	68.7	26.8	66.7	26.3	25.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.5	6.9	6.2	13.7	30.5	1.4	1.3	12.7	31.5	11.9	32.3	11.1	33.0	10.1	34.0		
(5)			12.7	23.8	1.3	0.9	11.8	24.4	11.0	24.9	10.2	25.4	9.3	26.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	20.1	23.3	24.2	23.5	21.5	19.5	18.2	17.6	17.0	17.3	18.1	19.4	21.3	(6)
(7)		DBStd	2.91	1.47	1.42	1.57	1.58	1.75	1.90	1.71	1.38	1.37	1.24	1.27	1.37	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	177	0	0	0	0	8	26	36	46	38	20	4	0	(9)
(10)		CDD10.0	3670	413	397	420	344	296	245	234	218	218	249	283	352	(10)
(11)		CDD18.3	804	155	163	162	94	44	21	12	6	6	11	37	93	(11)
(12)		CDH23.3	2342	543	753	652	182	42	16	9	3	3	2	13	123	(12)
(13)		CDH26.7	260	53	101	88	12	2	1	0	0	0	0	0	5	(13)
(14)	Wind	WSAvg	3.2	3.9	3.5	3.2	3.0	2.8	2.7	2.8	3.0	3.2	3.4	3.6	3.8	(14)
(15)	Precipitation	PrecAvg	10	1	1	0	1	0	1	1	2	2	0	1	1	(15)
(16)		PrecMax	16	3	3	4	1	2	3	4	4	5	1	2	2	(16)
(17)		PrecMin	5	0	0	0	0	0	0	0	0	0	0	0	0	(17)
(18)		PrecStd	3	1	1	1	0	0	1	1	1	1	0	1	1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.9	30.2	30.1	28.1	26.2	25.2	24.9	23.8	23.8	23.1	25.0	27.2	(19)
(20)			MCWB	23.8	23.8	23.5	22.0	20.6	20.2	19.6	19.4	19.3	18.5	19.8	22.0	(20)
(21)		2%	DB	28.0	29.0	28.8	26.2	24.2	23.1	22.1	21.2	21.2	21.9	23.2	25.7	(21)
(22)			MCWB	22.5	23.0	22.9	21.3	19.8	19.0	18.4	18.2	18.0	17.9	18.8	20.9	(22)
(23)		5%	DB	26.9	27.9	27.8	25.1	23.1	21.9	21.0	19.9	20.1	20.9	22.2	24.8	(23)
(24)			MCWB	22.1	22.6	22.5	20.8	19.3	18.8	17.7	16.8	17.1	17.3	18.3	20.4	(24)
(25)		10%	DB	26.0	27.0	26.5	24.2	22.1	20.9	19.8	18.8	19.1	20.0	21.8	23.9	(25)
(26)			MCWB	21.7	22.3	21.9	20.3	18.8	17.9	16.8	15.9	16.4	16.9	18.2	20.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.2	24.6	24.1	22.6	21.0	20.4	19.7	19.9	19.6	18.8	20.1	22.3	(27)
(28)			MCDWB	28.1	28.7	28.4	26.9	25.4	24.0	23.9	22.5	22.9	22.4	23.8	25.9	(28)
(29)		2%	WB	23.4	23.8	23.4	21.8	20.2	19.6	18.9	18.5	18.6	18.1	19.3	21.5	(29)
(30)			MCDWB	26.8	27.1	27.4	25.2	23.6	22.9	22.0	21.5	21.1	21.0	22.6	24.8	(30)
(31)		5%	WB	22.7	23.3	22.8	21.1	19.6	18.7	17.7	16.7	17.2	17.5	18.7	20.7	(31)
(32)			MCDWB	26.0	26.6	26.4	24.2	22.6	21.5	20.5	19.2	19.8	20.2	21.7	24.0	(32)
(33)		10%	WB	22.0	22.7	22.2	20.7	19.1	18.0	16.9	16.0	16.3	17.1	18.3	20.1	(33)
(34)			MCDWB	25.4	26.0	25.7	23.6	21.7	20.5	19.1	18.1	18.6	19.6	21.1	23.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.6	6.2	6.3	5.7	4.6	3.4	3.1	3.3	3.7	4.2	4.6	5.0	(35)
(36)			MCDBR	6.4	6.8	7.2	6.6	5.9	4.7	4.4	4.4	4.6	5.0	5.2	5.8	(36)
(37)			MCWBR	2.5	2.5	2.8	2.7	2.6	2.2	2.0	2.2	2.1	2.3	2.2	2.5	(37)
(38)		5% WB	MCDWB	6.0	6.2	6.7	6.1	5.6	4.5	3.9	3.7	4.4	4.7	4.9	5.5	(38)
(39)			MCWBR	2.3	2.4	2.7	2.6	2.5	2.1	1.9	2.0	2.2	2.3	2.3	2.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.499	0.490	0.510	0.502	0.448	0.448	0.426	0.449	0.535	0.486	0.479	0.488	(40)	
(41)		taud	2.186	2.228	2.168	2.162	2.282	2.266	2.344	2.259	1.999	2.170	2.201	2.199	(41)	
(42)		Ebn at Noon	853	856	822	795	808	791	821	829	789	849	865	861	(42)	
(43)		Edh at Noon	158	150	156	149	125	123	116	133	182	158	154	155	(43)	
(44)	All-Sky Solar Radiation	RadAvg	6.16	5.99	5.88	5.68	4.82	3.87	3.75	4.22	4.81	5.50	6.03	5.93	(44)	
(45)		RadStd	0.46	0.27	0.22	0.28	0.41	0.29	0.23	0.22	0.33	0.30	0.50	0.51	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+122
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page